



π -Conjugation

Begoña Milián-Medina and Johannes Gierschner*

In the current overview, the concept of π -conjugation in organic compounds is reexamined. Starting from valence bond and molecular orbital (MO) theory, the principal elements of conjugation are worked out starting from a simple Hückel picture; this includes the definition of the conjugated path, bond length alternation, as well as MO localization effects, and requires the distinction between formal and effective conjugation. The latter will be needed to understand the electronic and optical properties of polyconjugated molecules, highlighting the importance of a correct quantum chemical description of ground and excited state properties. Finally, implications of conjugation on interacting systems are shortly discussed. © 2011 John Wiley & Sons, Ltd.

How to cite this article:

WIREs Comput Mol Sci 2012, 2: 513–524 doi: 10.1002/wcms.95

INTRODUCTION

Polyconjugated organic materials play a major role in biology, chemistry, and material science because of their specific electronic, optical, and photophysical properties, all related to nature and extent of the conjugated path. For instance, in biological systems, the conjugation path determines the color of natural dyes, the phenomena of bioluminescence, as well as charge and energy transfer processes in photosynthesis. In material science, the versatility of organic synthesis provides tailor-made conjugated materials, e.g., see recent special issues on the application of conjugated materials for optoelectronic applications.^{1–4} Although the basic parameters on how to understand and tune the geometrical, electronic, and optical properties of conjugated materials are essentially known, simple design concepts toward materials with optimized conjugation patterns for devices are, however, difficult to establish because the change of one parameter to tune a specific property might be detrimental to others.^{1–3,5,6} Such precise requirements demand computer-aided approaches, which guide the synthetic process to design conjugated materials with the appropriate functionality. In fact, the rapid progress of computer technology has enabled a fast development of quantum chemical methods, where those based on density functional theory (DFT) play a key role because of the possibility of treating large systems

including electron correlation, at a reasonable computational cost. Much effort has been made to explore new functionals that can reproduce and predict specific properties.⁷ However, in the multitude of functionals and strategies, conceptual approaches tend to fade into the background, while 60 years ago this was the opposite. Indeed, the conjugation concept was introduced in the beginning of the 20th century with the emergence of synthetic chromophores to explain their (relative) chemical stability as well as the colors of organic dyes and pigments, which were early correlated to the presence of a high number of double bonds in their chemical structure.⁸

Thus, the present paper will relook at old ideas in the light of new methods. Because of the relevance in material science as well as to systematic reasons, we will concentrate our efforts on π -conjugated oligomers, which constitute a special class of organic dyes with a repetitive structure (characterized by a defined number of repetition units) as well as their polymer analogues, which are characterized by their molecular weight and polydispersity index; examples are shown in Figure 1. In any case, the reader will keep in mind that conjugation in general is a wider field, not only including π -conjugation but also σ -conjugation, σ - π -conjugation, and hyperconjugation and is also not limited to organic compounds. It is also discussed in alternating systems based on d-orbitals, although the reactivity of these classes of compounds is distinctively different. Moreover, there is a certain overlap of the π -conjugation concept with other relevant concepts such as (anti-)aromaticity. All these important aspects are separately covered in

*Correspondence to: johannes.gierschner@imdea.org

Madrid Institute of Advanced Studies, IMDEA Nanoscience, Campus Universitario de Cantoblanco, Madrid, Spain

DOI: 10.1002/wcms.95

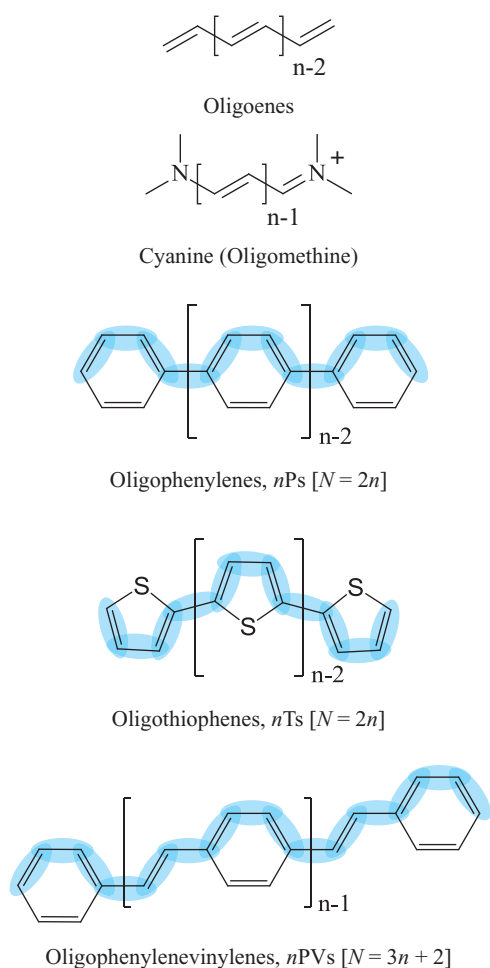


FIGURE 1 | Examples of conjugated oligomers where n is the number of repetition units and N is the conjugation length. The conjugation path is shown in blue.

specialized *WIREs Computational Molecular Science Review* articles.^{9,10}

We will start with an introduction to the concept of π -conjugation, based both on valence bond (VB) and molecular orbital (MO) theory. Herein, an example within the simple Hückel picture will provide us with the necessary ingredients to correctly describe conjugation, which are bond length alternation (BLA), the definition of the conjugated path, and MO localization, all discussed in the following sections. The section on MO localization will teach that a distinction between formal and effective conjugation has to be made in order to understand the observed phenomena. This will be especially important in the treatment of optical transitions, which will be discussed hereafter, also introducing excited state (ES) conjugation. In the final part, we will shortly discuss the importance of conjugation toward an un-

derstanding of interacting conjugated systems, including electronic, excitonic, and excimeric couplings, which govern charge transport, as well as optical and photophysical properties in molecular aggregates and crystals.

THE CONJUGATION CONCEPT

Conjugation is generally defined in a Lewis picture as the alternation of (formal) single and double (or triple) bonds along a chain of carbon atoms (or hetero atoms) providing free valences; accordingly, the *conjugation length* (N) might be defined as the number of double bonds along the shortest path between the terminal C atoms (Figure 1).¹¹ In this VB description, the molecular backbone is built by sp^2 -hybridized carbons, whereas the p-orbitals (being perpendicular to the sp^2 backbone) form alternating single and double bonds. Charged or radical mesomeric (resonance) structures are energetically disfavored; however, they give rise to a certain degree of delocalization. Their contribution will depend on the nature of the molecular backbone and the solvent; for charged polymethines (e.g., cyanines, see Figure 1), the resonance structures are energetically equivalent. VB theory is strong from its intuitive character and is helpful for chemists to understand chemical reactivity not observed in non-conjugated dienes as well as stability. In fact, 1,3-butadiene is 3.5 kcal/mol more stable than two independent ethylenes, being however a factor of 10 smaller than benzene as an example for an aromatic, i.e., ideally delocalized system.¹² This extra stability is known as *conjugation* (or delocalization) *energy* and it is determined on an energy basis by measuring the heat of hydrogenation or combustion. On a geometrical basis, conjugation might be defined through the BLA, *vide infra*. For the distinction of conjugated and aromatic materials, e.g., through nuclear magnetic resonance (NMR), see Ref 13.

Despite the strengths of VB theory, which even provides qualitative insight into ES properties,¹⁴ the need to quantitatively understand and predict electronic and optical properties of conjugated systems triggered the use of MO theory, a model which found entrance in modern quantum chemical methods. In the framework of MO theory, the explanation of conjugation is based on the electron delocalization over all atoms in a given MO. These MOs are formed through linear combination of atomic orbitals (LCAOs) with coefficients c_{ij} for AO i in MO j and gives as a result energies as well as sign and size of the LCAO coefficients (i.e., MO *topologies*) for a given system. The consequences of conjugation on the

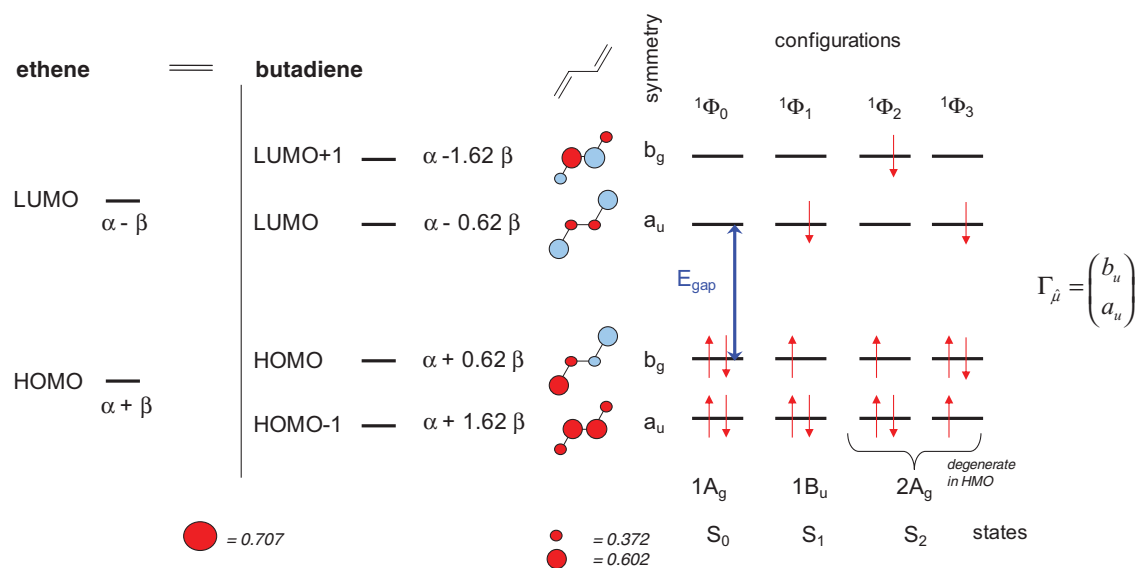


FIGURE 2 | Energy and topology of the frontier MOs in ethylene and butadiene, and electronic configurations (${}^n\Phi_i$) describing the ground (S_0), first (S_1), and second (S_2) singlet excited states of butadiene in the framework of HMO theory. The symmetry of the orbitals and the states is given in lower case and capital letters, respectively. $\Gamma_{\hat{\mu}}$ gives the symmetry elements of the transition dipole moment operator.

electronic structure are easily illustrated at a basic level by Hückel MO (HMO) theory.¹⁵ This one-electron approach concentrates solely on the (anti-)bonding π (π^*) type MOs, which are formed by adjacent carbon p-type AOs, whereas the sp^2 type AOs, which give rise to the σ (σ^*) MOs, are neglected. This separation can be done due to their orthogonality and the rather large energetic distance between σ and π (and σ^* and π^*) type MOs in many conjugated hydrocarbons, so that the first sets of (un)occupied frontier MOs, which determine the molecules' electronic and optical properties, are of π (π^*) type nature, see, e.g., in Figure 2 for the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) of ethylene.

As a simple conjugated system we might consider 1,3-butadiene formed from two ethylene molecules (Figure 2), being not only the smallest linear polyene but also simple in the sense that it belongs to the group of alternant hydrocarbons, which show a symmetrical energetic splitting of the respective (un)occupied MOs (HOMO- m , LUMO+ m , $m = 0, 1, 2$), with the same LCAO coefficients on each m ($c_{ij} = c_{ij'}$).¹⁶ This family of conjugated molecules comprises neutral open chains and some charged and radical analogues as well as (condensed) even-membered rings. In fact, for small and medium representatives, the HMO model is able to give a simple interpretation of many important phenomena¹⁷ such as optical properties,¹⁵ bond order, chemical sta-

bility (including stabilization of charged or radical species), specific reactivity¹⁸ (including Diels–Alder [2+4] cycloaddition and pericyclic photoreactions),¹⁹ substituent effects²⁰, and allows for the understanding of quinoid versus aromatic systems (via Hückel's rule)¹⁷ and the concept of (anti-)aromaticity²¹; for the relevance of the latter to material science, see, e.g., Ref 22. Even for many nonalternant systems, e.g., oligothiophenes (n Ts, Figure 1), HMO allows for some principle understanding.

The butadiene π (π^*) type MOs can be seen as the result of the interaction of the HOMO and LUMO of two ethylenes. As a consequence of this interaction, the HOMO (LUMO) of butadiene is higher (lower) in energy compared with those of ethylene, respectively, see Figure 2. This has a direct impact on the reactivity of the compounds. Comparison of the total energy of butadiene with ethylene gives an estimation of the *conjugation energy* of 0.48β at the HMO level. The MO topology provides further insight; in the HOMO of butadiene, the LCAO coefficients of the central carbon atoms are smaller compared with the terminal ones but show antibonding interaction due to the change of sign, whereas in the HOMO-1, the interaction is bonding with larger LCAO coefficients. This directly provides the explanation why the central bond represents an intermediate between a single and a double bond because the ground state (GS) properties are determined by the product of all occupied MOs; on more advanced quantum chemical calculations of the

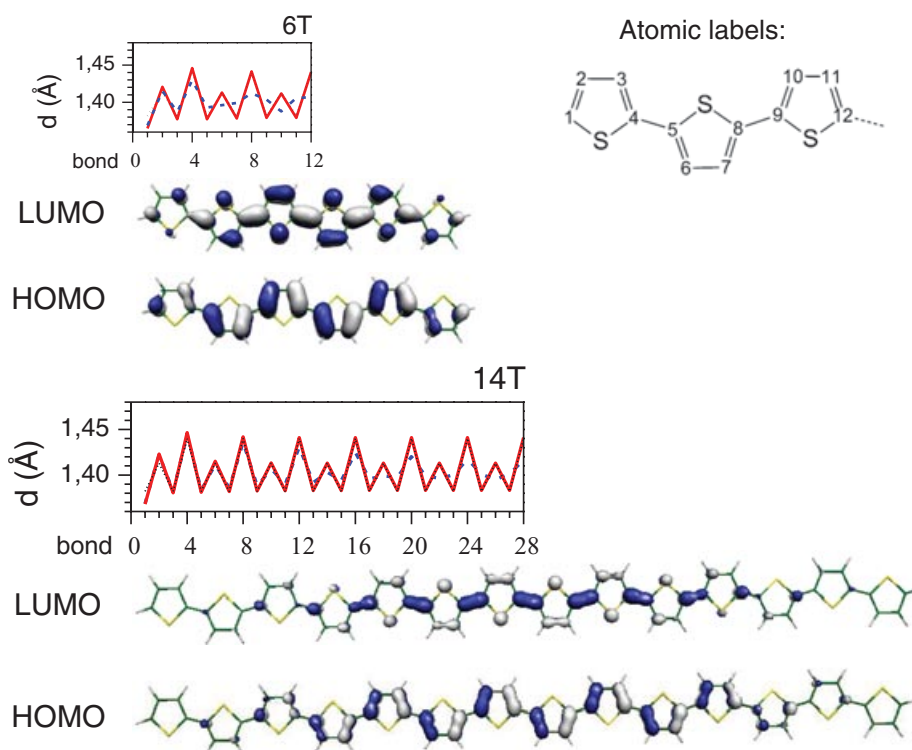


FIGURE 3 | Carbon–carbon bond lengths (d) in Å along the conjugated thiophene core (1 = terminal bond) of oligothiophenes 6T, 14T in the ground (solid red line), and first excited state (dashed blue) as calculated at the B3LYP/6-311G* level of theory; the bond pattern for ideal polythiophene in the ground state is given as thin dotted black line. Orbital pictures depict the HOMO (down) and LUMO topologies (up) of the n -ring oligomers.

conjugation energy and its determination, see the discussion in Refs 23 and 24.

HMO theory allows for a qualitative understanding of MO energies and topologies for a multitude of small and medium size molecules, however details of the MOs (energies and LCAO coefficients) depend sensitively on the theoretical methodology applied. This is especially true for longer conjugated systems because errors in the conjugation energies will rapidly accumulate upon elongation of the chain. The intrinsic HMO errors are obvious; geometric factors (bond lengths) are not included in the theoretical framework and delocalization is treated in a one-electron picture, which neglects long-range Coulomb-type interactions,¹⁷ thus leading to a systematic overestimation of the delocalization. Consequently, the HOMO–LUMO energy difference, which in HMO theory for linear polyenes reads $E_{HL} = 4\beta\cos\{\pi N/(2N+1)\}$, vanishes for an infinite chain in contradiction to experiment, *vide infra*.

Our small exercise in HMO theory gave some essential insight in the most important factors that govern conjugation, which are the (i) description of the conjugation pathway, (ii) electronic delocaliza-

tion, and (iii) BLA. In the following sections, we will discuss these issues in more detail before going to ES description of conjugated systems.

BOND LENGTH ALTERNATION

The failure of HMO for the description of infinite polyene-like conjugated systems demonstrates that a first measure for the reliability of theoretical methodologies is their ability to correctly describe the BLA,²⁵ generally given by the difference between the C–C single and double bonds in a conjugated system.²⁶ Accordingly, the degree of BLA in the GS can be used, e.g., to classify the systems into benzoid or quinoid, with the latter showing an inverted BLA pattern.^{26,27} In polyene-like periodic/infinite systems, the BLA is usually calculated as the largest difference δ_{BLA} between the carbon–carbon bond lengths. In finite systems, however, the bond lengths at the end parts differ from those in the middle part of the molecule and therefore giving a single BLA parameter might be sometimes misleading. In these cases, the use of BLA plots (Figure 3) is beneficial, showing the individual bond lengths against the site.²⁸ Finally, we will

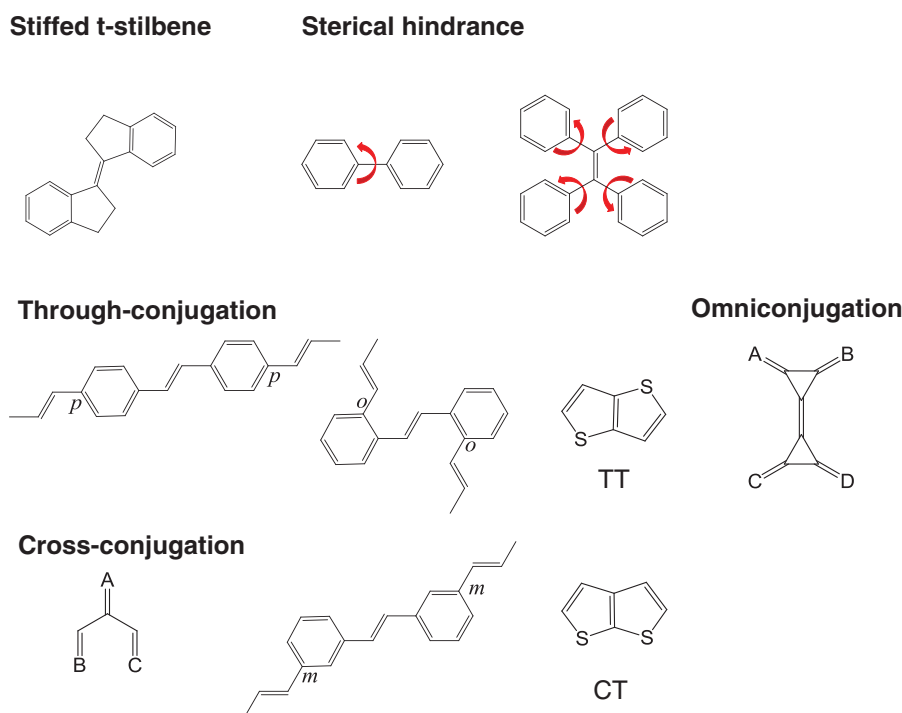


FIGURE 4 | Molecules with stiffed backbone, sterical hindrance, and *through- omni- and cross-conjugation*; p, m, and o indicate *para-*, *meta-* and *ortho-*substitution.

see later the need for a consideration of the BLA also for the ES.²⁹

The BLA is conceptually simple, however its accurate quantum chemical description is still challenging.^{25,30} In order to provide an experimental reference, precise bond length measurements are required; consequently, the size of the system plays a relevant role. Although very small systems can be experimentally measured using high precision techniques, e.g., rotational spectroscopy, and calculated by advanced quantum chemical methods, e.g., coupled-cluster with single and double and perturbative triple excitations CCSD(T), and CASSCF/CASPT2 (complete active space self-consistent field/with second-order perturbation theory), geometries of medium/larger-sized polyene-like systems have to rely on, e.g., X-ray analysis and less sophisticated quantum chemical tools. We might take a look, e.g., at *t*-stilbene (i.e., *n*PV with *n* = 1, see Figure 1) as an example of a medium-sized system (seven π -electron pairs, *N* = 5) and as an important building block in polyconjugated materials, which was intensively studied both experimentally (through gas phase electron diffraction and single-crystal X-ray diffraction)^{31,32} and by a multitude of theoretical methods. The latter range from cost-efficient semiempirical (e.g., AM1) and DFT^{33,34} to *ab initio* Hartree–

Fock (HF),^{29,35} post-HF (e.g., MP2),³³ and multiconfigurational methods (e.g., CASSCF).^{33,35} In the solid state, *t*-stilbene is planar, thus promoting conjugation of the vinylenic unit with the two phenyl groups. The BLA in the phenyl rings is very small, i.e., only slightly larger than in a perfectly delocalized system (benzene), however the vinyl single bond is about 1.472 Å, and the central double bond amounts 1.326–1.336 Å, see Table 1.^{31,32} The variability is ascribed to a temperature effect, originating from the torsional motion around the C-phenyl bonds. This was evidenced through a ‘stiff’ *t*-stilbene (Figure 4) where this torsion is suppressed, which gave 1.355 Å for the C=C bond, practically independent on temperature, and the vinyl single bonds are 1.476 Å.³⁷ The fact that torsions in conjugated systems are active even in the solid state under ambient conditions makes detailed comparisons for flexible conjugated systems with calculations, in general, difficult. For the ‘stiff’ *t*-stilbene, DFT(B3LYP) gives reliable results, see Table 1. For *t*-stilbene, B3LYP performs similar well to coupled-cluster ansatz with double excitations (CCD) and CASSCF, whereas HF tends to localize single and double bonds.³⁸

Upon further increase of the conjugated system, reliable experimental data become scarce. Thus, assessment of, e.g., different DFT functionals are

TABLE 1 | Geometry and Bond Length Alternation (δ_{BLA}) in Å of the Double and Single Bonds of the Vinylene Unit in *Trans*-Stilbene and Stiff-*Trans*-Stilbene at Different Computational Levels

Method basis set	AM1 ¹	HF ² 6-311G*	B3LYP 6-311G*	BHLYP 6-311G*	MP2 ³ 6-31G*	CCD ⁴ 6-31G*	CASSCF ² ANO	exp
<i>trans</i> -stilbene								
C=C	1.344	1.327	1.345 ²	1.331	1.352	1.345	1.350	1.326 ⁵ 1.336 ⁶
C—C	1.453	1.477	1.464 ²	1.463	1.465	1.476	1.476	1.471 ⁵ 1.472 ⁶
δ_{BLA}	0.109	0.150	0.119	0.132	0.113	0.131	0.126	0.145 0.136
<i>stiff-trans</i> -stilbene								
C=C			1.356					1.355 ⁷
C—C			1.474					1.476 ⁷
δ_{BLA}			0.118					0.121

¹Ref. 34.²Ref. 35.³Ref. 33.⁴Ref. 36.⁵X-ray data at 295 K, Ref. 32.⁶X-ray data at 113 K, Ref. 31.⁷X-ray data at 120 K, Ref. 35.

compared, for instance, against (spin-component-scaled) MP2 calculation as recently done on long polyenes²⁵; accordingly, medium HF exchange was suggested. A detailed look at longer systems is rather important because *end effects* are expected, which might sensitively influence conjugation, especially in the ES as we will see later. For the GS, Figure 3 compares the bond lengths of medium and long oligothiophenes (6T, 14T) as obtained at the B3LYP level. In fact, for both 6T and 14T, the BLA is very similar, being more pronounced in the terminal rings.

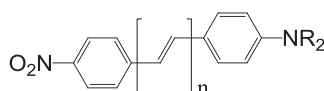
THE CONJUGATED PATH

In more complex molecules than linear polyenes, for instance, systems with branched conjugation pathways, the energy of the frontier orbitals will not only depend on the conjugation length (N) defined above but also on nature, number, and position of substituents with inductive ($\pm I$) or mesomeric ($\pm M$) effect as well as conjugated branches.¹¹ *Through-conjugation* provides a full π -conjugated pathway between all parts of the conjugated system. It can be found, e.g., in the oligomers of Figure 1 as well as their branched derivatives in *para*- and *ortho*- positions (Figure 4), including substituents with $\pm M$ effect, which in all cases significantly (de-)stabilize the LUMO (HOMO) positions compared with a polyene of the same N . As a special case, *omni-conjugation* allows for through-conjugation of many functional entities, see Figure 4.³⁹

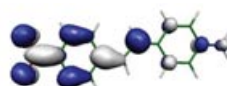
Cross-conjugation is applied to compounds in which two separated conjugated branches are linked to the same sp^2 -hybridized carbon atom,^{10,40} e.g., by *meta*-conjugation, see Figure 4, generating a break of the conjugation through a nodal plane. A further example is cross-conjugated thienothiophene (CT; Figure 4): introduction into a polymer chain leads to significantly lower HOMO values with respect to the through-conjugated counterpart TT, see Figure 4.⁴¹ In any case, electronic communication between cross-conjugated segments is observed; for details, see the Overview¹⁰ in *WIREs Computational Molecular Science*. In fact, even complete conjugation breaks found in systems with sp^3 -carbon defects in the conjugated chain, which confines the electronic excitations over small subunits acting as individual chromophores, still allow for some (through-bond and through-space) electronic communication between the segments.⁴²

Sterical effects can further significantly reduce the effective conjugation path, to which we will return below; examples are phenyl substituents in the vinylene unit of *t*-stilbene or biphenyl-based systems (see Figure 4), where the interring torsional angle is far from planar, not allowing for effective overlap between the p-type AOs. Consequently, the conjugation as observed from experiment is much smaller than expected for a planar system.¹¹ At ambient conditions, such distortion can be also induced through thermal population of low frequency modes in systems with planar equilibrium structures (e.g., *n*PVs

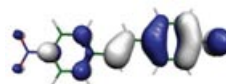
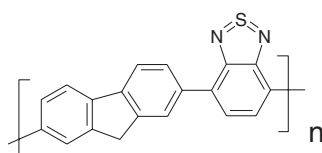
donor-acceptor structures

D- π -A

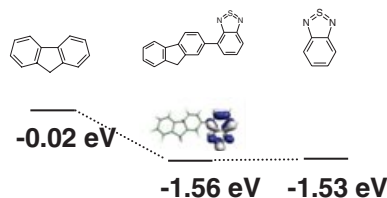
LUMO



HOMO

(DA)_n

LUMO



HOMO

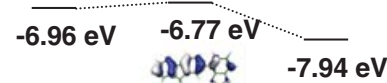


FIGURE 5 | Representative examples for donor–acceptor substituted conjugated molecules and their frontier MO topologies as calculated at the DFT(B3LYP/6-311G*) level of theory.

and n Ts, see Figure 1) due to the shallow torsional potentials around the single bonds in the electronic GS.^{29,43}

MO LOCALIZATION AND EFFECTIVE CONJUGATION

The continuous conjugation path as defined above is a necessary but not sufficient condition to provide efficient conjugation, thus an increase in N does not necessarily lead to a concurrent higher degree of the latter. This is due to MO localization effects, which can effectively limit conjugation. One factor that governs MO localization is the elongation of the conjugation length itself; Figure 3 compares the frontier MOs for 6T and 14T as calculated at the DFT(B3LYP) level. For 6T, the LCAO coefficients are rather equally distributed along the carbon backbone with just slightly smaller coefficients in the terminal rings, whereas in 14T, the LCAO coefficients are practically zero in the terminal rings. This is very different from the predictions of HMO theory, where the LCAO coefficient for atom i in the HOMO of linear polyenes reads $c_{i,\text{HOMO}} = \{2/(2N + 1)\}^{1/2} \sin\{i\pi N/(2N + 1)\}$, thus being large in the terminal positions also for long

chains. The difference is due to the neglect of long-range Coulomb-type interactions in HMO,¹⁷ and stresses the need for an accurate description of electron correlation in the quantum chemical methodology to correctly predict MO delocalization in conjugated systems. As MOs are not observables (in fact, observables are rather electron densities, e.g., from scanning tunneling microscopy, STM, experiments at different bias voltage), it is difficult to judge the ‘correct’ performance of the different methods on the matter and we will return to this point in the discussion of optical transitions.

Specific MO localization can be furthermore effectively introduced by donor–acceptor (DA) co-oligomers, where the acceptor LUMO is energetically stabilized against that of the donor. These structures can be either of repetitive (DA)_n or of D- π -A type, where π denotes a conjugated bridge (see Figure 5). For (DA)_n systems, at MO offsets between the D/A fragments larger than approximately 1.5–2.0 eV, complete spatial separation of the frontier MOs is observed.⁴⁴ For D- π -A systems, common frontier MOs are formed; however, the HOMO (LUMO) will be more localized on the donor (acceptor) unit despite through-conjugation in the system, see Figure 5.⁴⁵

In all, MO localization phenomena indicates the necessity to distinguish between *formal conjugation*, measured via the BLA as the product of the occupied MOs, and *effective conjugation*, governed by the MO topology of the frontier orbitals. The remarkable differences of topologies in the HOMO and LUMO of DA-type systems already indicate that promotion of electrons between such MOs, giving rise to optical transitions, will require more than the knowledge of the GS properties at which we looked so far.

ELECTRONIC EXCITATIONS

With the discussion on MO energies and topologies at hand, we are now able to tackle optical transitions, which as a start will be done again on butadiene in the HMO framework. In the one-electron picture of HMO, electronic excitations are simply obtained by the promotion of an electron from an occupied to an unoccupied MO defining new electronic configurations (Figure 2), so that the first ES is due to a HOMO→LUMO transition. The transition dipole moment $\tilde{\mu}_{if}$ is calculated from the respective wavefunctions ψ_i, ψ_f in the initial and final states through application of the dipole moment operator $\hat{\mu}$:

$$\tilde{\mu}_{if} = \int \psi_i^* \hat{\mu} \psi_f d\tau \neq 0 \quad (1)$$

The principle question whether such transition is allowed can be simply answered by symmetry consideration in the given point group of the molecule, i.e., the symmetry operation $\Gamma = \Gamma_i \Gamma_\mu \Gamma_f$ must contain the totally symmetrical element. For *trans*-butadiene (point group C_{2h}), HOMO→LUMO is allowed, whereas the second transition (degenerate in HMO due to equal HOMO−1→LUMO and HOMO→LUMO+1 energies) is symmetry forbidden, see Figure 2; for the *cis*-isomer (C_{2v}), both transitions are allowed. Polarization and (relative) strength of the transition are finally calculated through Eq. (1) by multiplication of the LCAO coefficients for each atom in the respective initial and final MOs to give the transition dipole density (TDD), from which $\tilde{\mu}_{if}$ is formed through summing up the individual vectors.

This qualitative picture is not changed by using more sophisticated methods; however, energy, intensity (oscillator strength), polarization, and composition (i.e., configuration interaction or CI description) of the electronic transition will depend sensitively on the method and thus on the correct description of conjugation in the system. In order to compare calculated optical transitions with experiment, a number of steps have to be added to the simple scheme

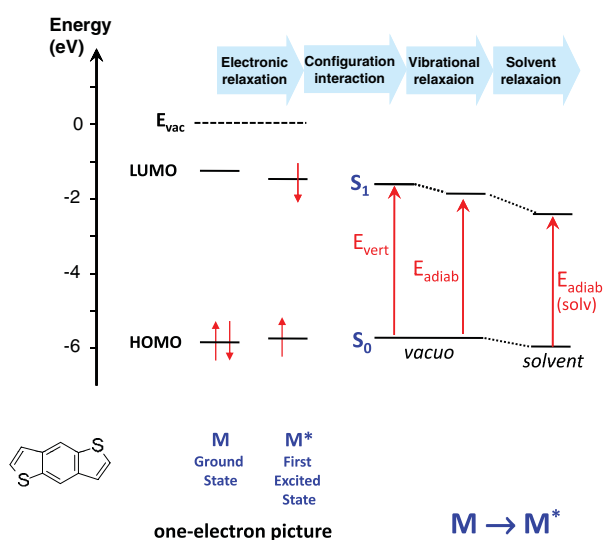


FIGURE 6 | Electronic excitation for benzodithiophene as calculated at the (TD-)DFT(B3LYP/6-311G*) level of theory.

above, see Figure 6. After promotion of an electron from the HOMO to the LUMO, the system will electronically relax. Then, CI is added to define the excited (Franck–Condon) state (vertical transition), which vibrationally relaxes toward the minimum of the ES potential hypersurface (relaxation or equilibration energy) to give the adiabatic transition (optical bandgap). Finally, solvent effects have to be included to accurately reproduce the experiment; for a detailed comparison between experiment and theory in a recent review, see Ref 11. Hence, the applied quantum chemical method has to correctly reproduce all steps, which are each intimately correlated to the *changes in the conjugation* upon electronic excitation. We will limit the discussion to conjugated systems with a symmetry-allowed first-excited singlet state (S_1), e.g., *n*Ts and *n*PVs, where the state can be described in a good approximation by a HOMO→LUMO excitation (with usually more than 95% participation), covering the main absorption features and being responsible for the emission (fluorescence) process.

In a VB picture, the energy of an optical transition is correlated with the change of the BLA upon electronic excitation, which can be accessed by ES optimization. For *n*Ts as calculated by (TD-)DFT (B3LYP, Figure 3), the BLA decreases from the benzoid structure in the electronic GS (S_0) with large δ_{BLA} toward a more delocalized structure in S_1 with small δ_{BLA} in the centre of the molecule, whereas in the extremities, the benzoid structure is retained. The size of the undistorted terminal parts of the molecule increases with chain length as can be seen by comparing

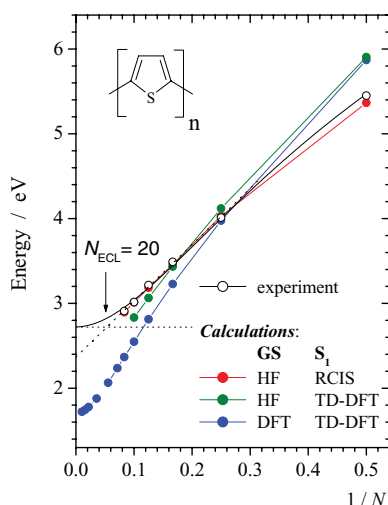


FIGURE 7 | Vertical transition energies *in vacuo* for oligothiophenes as function of $1/N$; experiment (open symbols, Ref 43), calculations with different combinations of ground state optimizations with time-dependent calculations; (TD)-DFT calculations were done at the B3LYP level, the 6-311G* basis set was used in all cases. The ECL is extracted graphically by the auxiliary (dashed) lines, see Ref 11.

6T and 14T, thus effectively limiting the central part of the molecule where a change of the BLA upon electronic excitation is observed. This behavior in S_1 results from the interaction between the now half-filled HOMO and LUMO orbitals, which are both localized toward the centre of the molecule with vanishing LCAO coefficients at the extremes, see Figure 3. This is the reason for the moderate optical bandgap experimentally observed in these systems.⁴³

The B3LYP results are qualitatively reproduced by other methods, where in restricted CI singles calculation on a HF optimized geometry (RCIS//HF) leads (as expected) to somewhat stronger MO localization. Small differences in MO localization might lead to systematic deviations in the chain length evolution, which can be conveniently evaluated by plotting the calculated transition energies against $1/N$ and comparing them to experiment, see Figure 7. Detailed studies on different classes of oligomers in the past showed that standard DFT functionals with small to medium HF exchange lead to a systematic overestimation of the slope in the $1/N$ plot when combining DFT-optimized geometries with TD-DFT, whereas the size-consistent treatment in the RCIS//HF framework leads to a good agreement with experiment, as long as the ES is symmetry allowed (high oscillator strength) and fully geometry optimized at the RCIS level.^{11,28} This indicates that the MO localization is rather well described within this framework.

Geometrical factors (i.e., the precise description of the BLA in the GS) on the other side do not play a prominent role; the calculated vertical transition energies by TDDFT(B3LYP)//DFT(B3LYP) hardly change when based on HF geometries, i.e., TDDFT(B3LYP)//HF, see Figure 7. This stresses the importance of a correct MO description as well as the proper calculation of electronic and geometrical relaxation. The latter can be explicitly calculated through the projection of the geometrical change upon electronic excitation onto the normal modes of vibrations.^{29,46} As the relative change in the BLA for $S_0 \rightarrow S_1$ decreases with N , the relaxation energy should decrease as well, in agreement with experiment.^{29,43}

The $1/N$ plots allow for the extrapolation of oligomer transition energies to the infinite conjugated polymer.¹¹ Very different to the predictions of HMO or the free electron model of H. Kuhn in its basic form,⁴⁷ polyene-like systems not only give a non-zero bandgap, which results from the inclusion of BLA as seen above, but also saturation above a certain chain length toward the polymer limit, the ‘effective conjugation length’ (ECL),⁴⁵ which originates from the localization of the excitation as a result of MO localization. The ECL can be graphically extracted from the $1/N$ plot (see Figure 7) and amounts to about $N \simeq 20$, see Ref. 11; for polyenes with incorporated triple bonds, the ECL is significantly shortened due to reduced π - π^* orbital interactions, despite larger delocalization energy as obtained from natural bond orbital analysis.⁴⁸ On a simple physical basis (which works reasonably well for a multitude of materials), the saturation of the excitation energies toward the polymer limit can be described through the coupling of a collinear array of harmonic oscillators as early done by W. Kuhn; for details, see Ref 11. For short polymethines (e.g., cyanines, Figure 1), linear extrapolation gives a zero bandgap,⁴⁹ due to equivalent resonant structures, reflected in vanishing BLA. In any case, for larger chain lengths, i.e., beyond the ‘cyanine limit’ at around $N \approx 10$, symmetry breaking localizes the MOs and limits the effective conjugation.⁴⁹

Localization of frontier MOs in different parts of the molecule through the above-mentioned DA type through-conjugated systems, i.e., $(DA)_n$ and $D-\pi-A$, may lead to excitations with (partial) intramolecular-charge-transfer (ICT) character, resulting in unstructured bathochromically shifted absorption bands of low oscillator strength as a result of the reduced overlap of the constituting MOs. In $D-\pi-A$ systems, the ICT character decreases with the length of the π -system, which leads to an

hypsochromic shift of the absorption band with N for short chains, contrary to the usual behavior of polyene-like systems discussed above.^{11,45} In $(DA)_n$ systems, the energies of localized versus delocalized MOs depend sensitively on the conjugation length as well as on the nature of the D and A moieties, which determine whether low-lying ICT transitions are observed.⁴⁴

The control of MO localization in through-conjugated systems, and concomitantly of the ES, might allow to further develop specific computational-based design concepts for tailor-made compounds with controlled properties, e.g., broad-band low bandgap materials,^{1–3,5} etc.

INTERACTING CONJUGATED SYSTEMS

Although conjugation has its most obvious primary impact in the electronic and optical properties of the conjugated molecules themselves as discussed above, we should shortly address a secondary effect arising from the interaction of conjugated systems in the aggregated state, e.g., in thin (self-assembled) films, single crystal etc., which involves static dipole interactions⁵⁰ as well as electronic and excitonic couplings. Because of the limited space in the present overview, we will simply describe some principal elements.

At the electronic level, such intermolecular interaction governs charge transfer, where the strength of the interaction between the HOMO (LUMO) orbitals is decisive on the efficiency of hole (electron) transport. A major transport parameter, the *electronic coupling* which is determined by the transfer integrals (calculated in a one-electron picture by half of the splitting of the frontier MOs through the interaction), depends sensitively on the exact mutual position between the MOs of the interacting conjugated molecules in space as well as their MO topology, so that the control of conjugation is a major step in the design of improved charge transport materials.⁵¹

Also the optical and photophysical properties in the solid state are strongly influenced by the interactions of the conjugated molecules, this time however triggered by the *excitonic coupling* of the molecular TDDs, as formed from the respective MOs in the given molecular electronic transition. Depending on the exact mutual arrangement, the absorp-

tion spectra suffer from spectral shifts against the isolated molecule as early described in the ‘molecular exciton model’ of Kasha,⁵² which might be hypsochromic (in so-called H-type aggregates) or bathochromic (J-aggregates) with decreased (H) or increased (J) radiative rates. Quantum chemical formulation of the excitonic interaction through the TDD^{4,53} allows for detailed computational simulation of the optical spectra.⁵⁴ The magnitude of the effect will furthermore depend on the spatial extent and nature of the TDD, and thus on the precise effective conjugation of the system. Different to the electronic coupling, which experiences only the interaction of adjacent molecules, excitonic coupling extends far beyond the ‘nearest neighbor approximation’,⁵³ and shows a peculiar peak behavior with increasing conjugation length, so that medium size systems show the strongest excitonic coupling, whereas the latter goes to zero for infinite polymer chains, being a direct consequence of the localized ESs in neighboring chains.⁵³ Finally, conjugation also impacts the solid-state absorption and emission band shapes in π -stacked systems, due to different interactions of the conjugated systems in the GS and ES, which give rise to a change in the intermolecular separation upon electronic excitation and thus to ‘excimer-like’ bandshapes.⁵⁵ A correct theoretical description of π -stacked systems at the DFT level requires dispersion-corrected functionals, as recently reviewed.⁵⁶

CONCLUSION

The present overview on conjugation looked at old ideas in the light of new methods. Starting with conceptual approaches based on VB and MO theory, we distilled the principal elements of conjugation, including the description of the conjugated path, BLA in the GS and ES, and MO localization phenomena. The latter indicated the necessity to distinguish between formal conjugation (measured via the BLA) and effective conjugation (governed by the MO topology) and provided the tools to understand the consequences on the electronic and (linear) optical properties of conjugated oligomers, highlighting the importance of a correct description by quantum chemical methods. In the last part, interacting conjugated systems were briefly discussed, addressing the impact of conjugation on intermolecular electronic and excitonic coupling.

ACKNOWLEDGMENTS

We are grateful to our colleagues E. Ortí, J. Sánchez (Valencia), J. C. Sancho-García (Alicante), L. Lüer (Madrid), and H.-J. Egelhaaf (Nürnberg) for stimulating discussions. Johannes Gierschner is a Ramón y Cajal Fellow of the Spanish Ministry for Science and Innovation.

REFERENCES

1. Special issue on Organic electronics and optoelectronics. *Chem Rev* 2007, 107:923–1386.
2. Special issue on Materials for electronics. *Chem Rev* 2010, 110:1–574.
3. Special issue on Organic photovoltaics. *Acc Chem Res* 2009, 42:1689–1857.
4. Brédas JL, Beljonne D, Coropceanu V, Cornil J. Charge-transfer and energy-transfer processes in π -conjugated oligomers and polymers: a molecular picture. *Chem Rev* 2004, 104:4971–5004.
5. Beaujuge PM, Amb CM, Reynolds JR. Spectral engineering in π -conjugated polymers with intramolecular donor–acceptor interactions. *Acc Chem Res* 2010, 43:1396–1407.
6. Hains AW, Liang Z, Woodhouse MA, Gregg BA. Molecular semiconductors in organic photovoltaic cells. *Chem Rev* 2010, 110:6689–6735.
7. Zhao Y, Truhlar DG. Density functionals with broad applicability in chemistry. *Acc Chem Res* 2008, 41:157–167.
8. Lewis GN, Calvin M. The color of organic substances. *Chem Rev* 1939, 25:273–328.
9. Alabugin IV, Gilmore KM, Peterson PW. Hyperconjugation. *WIREs Comput Mol Sci* 2011, 1:109–141.
10. Limacher PA, Lüthi HP. Cross-conjugation. *WIREs Comput Mol Sci* 2011, 1:477–486.
11. Gierschner J, Cornil J, Egelhaaf HJ. Optical bandgaps of π -conjugated organic materials at the polymer limit: experiment and theory. *Adv Mater* 2007, 19:173–191.
12. Wiberg KB, Nakaji DY, Morgan KM. Heat of hydrogenation of a *cis* imine. An experimental and theoretical study. *J Am Chem Soc* 1993, 115:3527–3532.
13. Chen Z, Wannere CS, Corminboeuf C, Puchta R, Schleyer PvR. Nucleus-independent chemical shifts (NICS) as an aromaticity criterion. *Chem Rev* 2005, 105:3842–3888.
14. Shaik S, Hiberty PC. A primer on qualitative valence bond theory—A theory coming of age. *WIREs Comput Mol Sci* 2011, 1:18–29.
15. Heilbronner E, Bock H. *The HMO-Model and its Application*. Weinheim, Germany: VCH, GmbH; 1976.
16. Pariser R. Theory of the electronic spectra and structure of the polyacenes and of alternant hydrocarbons. *J Chem Phys* 1956, 24:250–268.
17. Kutzelnigg W. What i like about Hückel theory. *J Comput Chem* 2007, 28:25–34.
18. Dewar MJS, Longuet-Higgins HC. The correspondence between the resonance and molecular orbital theories. *Proc Roy Soc London A* 1952, 214:482–493.
19. Woodward RB, Hoffmann R. Stereochemistry of electrocyclic reactions. *J Am Chem Soc* 1965, 87:395–397.
20. Milián-Medina B, Beljonne D, Egelhaaf HJ, Gierschner J. Effect of fluorination on the electronic structure and optical excitations of π -conjugated molecules. *J Chem Phys* 2007, 126:111101/1–111101/6.
21. Minkin VI, Glukhovtsev MN, Simkin, BY. *Aromaticity and Antiaromaticity*. New York: Wiley; 1994.
22. Kertesz M, Choi CH, Yang S. Conjugated polymers and aromaticity. *Chem Rev* 2005, 105:3448–3481.
23. Zielinski M, Havenith RWA, Jenneskens LW, van Lenthe JH. A comparison of approaches to estimate the resonance energy. *Theor Chem Acc* 2010, 127:19–25.
24. Mo Y, Hiberty PC, von Ragué Schleyer P. How to properly compute the resonance energy within the *ab initio* valence bond theory: a response to the ZHJVL paper. *Theor Chem Acc* 2010, 127:27–38.
25. Sancho-García JC, Pérez-Jiménez AJ. Improved accuracy with medium cost computational methods for the evaluation of bond length alternation of increasingly long oligoacetylenes. *Phys Chem Chem Phys* 2007, 9:5874–5879.
26. Kertesz M. Bond length alternation and energy gap in $(\text{CH})_x$. Application of the intermediate exciton formalism. *Chem Phys* 1979, 44:349–356.
27. Casado J, Miller LL, Mann KR, Pappenfus TM, Higuchi H, Ortí E, Milián B, Pou-Amérigo R, Hernández V, López Navarrete JT. Quinonoid oligothiophenes as electron-donor and electron-acceptor materials. A spectroelectrochemical and theoretical study. *J Am Chem Soc* 2002, 124:12380–12388.
28. Tavan P, Schulten K. Electronic excitations in finite and infinite polyenes. *Phys Rev B* 1987, 36:4337–4358.

29. Gierschner J, Mack HG, Lüer L, Oelkrug D. Fluorescence and absorption spectra of oligophenylenevinyls: vibronic coupling, band shapes and solvatochromism. *J Chem Phys* 2002, 116: 8596–8609.
30. Jacquemin D, Adamo C. Bond length alternation of conjugated oligomers: wave function and DFT benchmarks. *J Chem Theory Comput* 2011, 7:369–376.
31. Hoeckstra A, Meerteens P, Vos A. Refinement of the crystal structure of *trans*-stilbene (TSB). The molecular structure in the crystalline and gaseous phases. *Acta Crystallogr Sect B* 1975, 31:2813–2817.
32. Bouwstra JA, Shouten A, Kroon J. Structural studies of the system *trans*-azobenzene/*trans*-stilbene. II. A reinvestigation of the disorder in the crystal structure of *trans*-stilbene, C₁₄H₁₂. *Acta Crystallogr Sect C* 1984, 40:428–431.
33. Choi CH, Kertesz M. Conformational information from vibrational spectra of styrene, *trans*-stilbene, and *cis*-stilbene. *J Phys Chem A* 1997, 101:3823–3831.
34. Galvao DS, Soos ZG, Ramasesha S, Eternad S. A parametric method 3 (PM3) study of *trans*-stilbene. *J Chem Phys* 1993, 98:3016–3021.
35. Gagliardi L, Orlandi G, Molina V, Malmqvist PA, Roos B. Theoretical study of the lowest ¹B_U states of *trans*-stilbene. *J Phys Chem A* 2002, 106:7355–7361.
36. Kwasniewski SP, Claes L, François J-P, Deleuze MS. High level theoretical study of the structure and rotational barriers of *trans*-stilbene. *J Chem Phys* 2003, 118:7823–7836.
37. Ogawa K, Harada J, Tomoda S. Unusually short ethylene bond and large amplitude torsional motion of (E)-stilbenes in crystals. X-ray crystallographic study of 'stiff' stilbenes. *Acta Cryst B* 1995, 51:240–248.
38. Choi CH, Kertesz M, Karpfen A. The effects of electron correlation on the degree of bond alternation and electronic structure of oligomers of polyacetylene. *J Chem Phys* 1997, 107:6712–6721.
39. Van der Veen MH, Rispens MT, Jonkman HT, Hummelen JC. Molecules with linear π -conjugated pathways between all substituents: omniconjugation. *Adv Funct Mater* 2004, 14:215–223.
40. Gholami M, Tykwinski RR. Oligomeric and polymeric systems with a cross-conjugated π -framework. *Chem Rev* 2006, 106:4997–5027.
41. Milián-Medina B, Van Vooren A, Brocorens P, Gierschner J, Shkunov M, Heeney M, McCulloch I, Lazzaroni R, Cornil J. Electronic structure and charge-transport properties of polythiophene chains containing thienothiophene units: a joint experimental and theoretical study. *Chem Mater* 2007, 19:4949–4956.
42. Hennebicq E, Deleener C, Brédas JL, Scholes GD, Beljonne D. Chromophores in phenylenevinylene-based conjugated polymers: role of conformational kinks and chemical defects. *J Chem Phys* 2006, 125:054901–054916.
43. Gierschner J, Egelhaaf HJ, Mack HG, Schweizer S, Doser B, Oelkrug D. Optical spectra of oligothiophenes: vibronic states, torsional motions, and solvent shifts. *Synth Met* 2003, 138:311–315.
44. Dutta T, Woody KB, Parkin SR, Watson MD, Gierschner J. Conjugated polymers with large effective Stokes shift: benzobisdioxole-based poly(phenyleneethynylene)s. *J Am Chem Soc* 2009, 131:17321–17327.
45. Meier H. Conjugated oligomers with terminal donor-acceptor substitution. *Angew Chem Int Ed* 2005, 44:2482–2506.
46. Birks JB. *Organic Molecular Photophysics*. London, UK, John Wiley & Sons; 1973.
47. Kuhn H. Neuere untersuchungen über das elektronen-gasmodell organischer farbstoffe. *Angew Chem* 1958, 71:93–101.
48. Bruschi M, Limacher PA, Hutter J, Luthi HP. A scheme for the evaluation of electron delocalization and conjugation efficiency in linearly π -conjugated systems. *J Chem Theory Comput* 2009, 5:506–514.
49. Tolbert LM, Zhao X. Beyond the cyanine limit: Peierls distortion and symmetry collapse in a polymethine dye. *J Am Chem Soc* 1997, 119:3253–3258.
50. Natan A, Zidon Y, Shapira Y, Kronik L. Cooperative effects and dipole formation at semiconductor and self-assembled-monolayer interfaces. *Phys Rev B* 2006, 73:193310.
51. Coropceanu V, Cornil J, da Silva Filho DA, Olivier Y, Silbey R, Brédas JL. Charge transport in organic semiconductors. *Chem Rev* 2007, 107:926–952.
52. Kasha M. Energy transfer mechanism and the molecular exciton model for molecular aggregates. *Radiat Res* 1963, 20:55–71.
53. Gierschner J, Huang Y-S, Van Averbek B, Cornil J, Friend RH, Beljonne D. Excitonic versus electronic couplings in molecular assemblies: the importance of non-nearest neighbour interactions. *J Chem Phys* 2009, 130:044105/1–044105/6.
54. Spano F. The spectral signatures of Frenkel Polarons in H- and J-aggregates. *Acc Chem Res* 2010, 43:429–439.
55. Gierschner J, Ehni M, Egelhaaf HJ, Milián Medina B, Beljonne D, Benmansour H, Bazan GC. Solid state optical properties of linear polyconjugated molecules: π -stack contra herringbone. *J Chem Phys* 2005, 123:144914/1–144914/9.
56. Grimme S. Density functional theory with London dispersion corrections. *WIREs Comput Mol Sci* 2011, 1:211–228.